

Article

Do Forest Experience, Socialization and Demographic Characteristics Affect the Attitudes toward Hunting of Youths from Urban Areas?

Hubert Codrow ¹, Adrian Łukowski ¹ , Michał Klimkiewicz ¹, Małgorzata Krokowska-Paluszak ², Anna Wierzbicka ^{1,*}  and Maciej Skorupski ¹

¹ Faculty of Forestry and Wood Science, Poznań University of Life Sciences, 60-625 Poznań, Poland

² Faculty of Agricultural and Forestry, University of Warmia and Mazury in Olsztyn, 10-719 Olsztyn, Poland

* Correspondence: anna.wierzbicka@up.poznan.pl

Abstract: Acceptance of forest management and its park, game and wildlife management is decreasing in Europe. Building a positive attitude toward game and wildlife management is a field of work of forest educators. To design and conduct effective activities, it was crucial to identify the specific needs. We conducted a survey among Polish high school students from cities and towns (1947 individuals) to find out answers to the following questions: What kind of attitude toward hunting (ATH) do Polish teenagers have? What shapes their ATH? Is this attitude and its drivers similar to the ones of adults? Half of Polish urban teenagers recognized that hunting in Poland is necessary. From socio-demographic factors, forest and hunting experience had the largest impact on teenagers' attitude toward hunting. Gender and social network had a smaller but still significant impact. Other factors, e.g., place of residence, had no impact on the ATH. Our results show that forest education should be focused on all young residents, no matter if they are from a big city or a small town, as teenagers' ATH is the same. Field trips and other active methods are recommended because personal experience has the largest impact on shaping ATH.

Keywords: attitude toward hunting; game management; teenagers; young adults; Poland



Citation: Codrow, H.; Łukowski, A.; Klimkiewicz, M.; Krokowska-Paluszak, M.; Wierzbicka, A.; Skorupski, M. Do Forest Experience, Socialization and Demographic Characteristics Affect the Attitudes toward Hunting of Youths from Urban Areas? *Forests* **2022**, *13*, 1803. <https://doi.org/10.3390/f13111803>

Academic Editors: Mika Rekola, Alexia Barrable and Alexios Arvanitis

Received: 14 September 2022

Accepted: 26 October 2022

Published: 29 October 2022

Publisher's Note: MDPI stays neutral with regard to jurisdictional claims in published maps and institutional affiliations.



Copyright: © 2022 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

Forest education or forest pedagogy is a wide term, depending on the country or even continent, its definition can differ. It generally includes knowledge about forest ecosystems and forest management. In most countries, part of it is also wildlife management and hunting as a segment of sustainable forestry [1]. To achieve the best possible effectiveness of education, assessment of the current state of knowledge and attitudes are vital. When it comes to attitudes and knowledge about wildlife management and hunting, most researchers focus on mature adults, and there is little research about teenagers and young adults [2]. However, identification of wildlife management and hunting is crucial to effective education. Acceptance of game management and hunting by people living in urbanized areas is lower than among people living in the countryside [3–9]. These relationships are not connected with age, with the same attitudes being observed for adults, adolescents, and young adults. It has been found that adolescents and young adults living in cities are generally less agreeable to hunting than their peers living in the countryside [5,10–12]. Additionally, Hauser [9] noticed that city dwellers who had moved from the countryside have a more favorable attitude toward hunting than those who have lived in cities all their lives [13–17].

Brämer [18] states that the current generation of young people usually lives separated from nature. At the same time, these people have a patronizing and infantile nature, mainly seeing threats from humanity. Often these threats are exaggerated and sometimes even completely unrealistic. According to 75% of all teenagers, mankind is nature's worst enemy.

In the case of high school students, this percentage is even greater, and this view is shared by as many as 90% of them. This phenomenon seems to mainly concern inhabitants of large urban agglomerations, whose knowledge of nature comes largely from television and the Internet [18,19]. People, especially young people, are receiving information mostly from television and the Internet, which is very popular because currently, it is the most widespread mass media, with social media being freely available [20]. However, in the case of human–nature relations, television still plays a dominant role [21–23]. In the case of information on nature, this applies especially to young people living in cities. Young people are spending less time outdoors more time in front of a computer screen and further away from green areas [24]. The relationship that children and adolescents have with nature is a key foundation of what their attitudes concerning environmental and nature protection issues will be in the future [25–28], and this relationship is currently one of the main concerns of forest education.

In the case of attitudes toward hunting, social background is also important. According to Skogen [5], a teenager from a well-educated family who moved to the countryside, but whose parents commute to work in the city, typically has a negative attitude toward hunting. In contrast, teenagers brought up in a farming or working-class family have a positive attitude toward hunting. These conclusions partially overlap with the research by Brämer [18] on social origin and fit into the broader context of research on the attitudes of indigenous and immigrant villagers to the management of natural resources, indicating a conflict between the expectations and needs of these social groups in relation to the use of natural resources [29–32].

Personal experience is very important in shaping attitudes, especially toward nature and its management, which is why modern forest education stresses active methods [16,33,34]. In many countries, e.g., Finland, Great Britain, the USA, and Canada, teenagers can hunt and can accompany adults on hunting trips. On the other hand, Poland has forbidden children under 18 years old to hunt since 2018 [35]. This can result in a lower acceptance of hunting and game management and forest management in general in the future. So, it is vital to determine Polish young people's knowledge and attitudes toward game management to plan and shape forest education activities according to their needs. We conducted a survey to find the answers to the following questions:

- (1) What kind of attitude toward hunting (ATH) do Polish teenagers have?
- (2) What shapes their ATH?
- (3) Is this attitude and its drivers similar to that of adults?

2. Materials and Methods

2.1. The Questionnaire

A sample of 1947 young Polish citizens was gathered in 2020. Respondents filled out an online questionnaire during their school hours ($n = 1947$). Samples were collected during the whole month of May. We asked school authorities in high schools in four different places in Poland to send questionnaires to randomly chosen students (cities: Poznań, Wrocław, both around 600,000 inhabitants; Konin, Krotoszyn, Szczecinek, and Trzcianka-county “powiat” capitals, number of inhabitants between 30 to 70,000). Responders were asked to provide information regarding their age (14–16 years, 17–19 years, >20 years), gender (female, male), and residency (city, town). Respondents also provided information regarding their frequency of visits to urban and managed forests during the month prior to the survey (more than once, once, not once). They were questioned on whether their parents grew up in urban areas or in the countryside, or if one of the parents was raised in an urban area and the other in the countryside. In addition, they were asked if they knew any hunter or forest worker from their surroundings (yes, no). Lastly, respondents were asked if they would like to participate in a hunt as an observer to see what it really looks like and whether in the future they would like to become a hunter (yes, no, no opinion). Respondents' feedback was used in data analysis only if they had filled in all required points in the survey.

In total, we conducted a survey of 1947 respondents, none of which were excluded in further analysis because they had filled in all points of the survey.

2.2. Data Analysis

Results were merged based on seven statements determining the respondents' attitudes toward hunters and hunting (Table 1). We used Principal Component Analysis to assess the construct validity of items and Cronbach's α to measure the internal consistency [36]. To allow the use of Likert's five-point scale, respondents could choose from: I agree completely (2), agree somewhat (1), no opinion (0), disagree somewhat (−1), and disagree completely (−2). The setting scale ranged from −14 to 14.

Table 1. Opinions of young Polish respondents about seven items related to attitudes regarding hunting and hunters. Loadings (from Principal Component Analysis) of each item on principal component one (PC1) are also shown. Questions in bold specifically investigated negative attitudes regarding hunting and hunters (thus “I agree completely”, “agree somewhat”, “disagree somewhat”, and “disagree completely” were awarded −2, −1, 1, and 2 points, respectively). Data were collected in 2020 from a random sample of 1947 Polish high school students.

	Items	N1	% Agree	N2	% Disagree	N3	% No Opinion	Factor Loading PC1
1	I associate modern hunting in Poland with the necessary regulation of the number of game animals.	988	51	452	23	507	26	0.61
2	Populations of wild animals in Poland are currently at a stable level and do not require human regulation.	655	34	829	42	463	24	0.40
3	There are fewer and fewer wild animals in Poland, because hunters hunt them (deer, roe deer, fallow deer, and wild boar).	417	21	1226	63	304	16	0.68
4	Too little information is communicated by the hunters themselves about what they do.	1296	67	182	9	469	24	−0.21
5	I agree to kill wild animals (by hunters) for meat.	741	38	946	49	260	13	0.74
6	The media in Poland show hunting in an objective way.	662	51	491	23	794	26	0.09
7	I would feel comfortable because it is a social group that I trust.	616	32	710	36	621	32	0.76

Before starting the statistical analysis of the results, the initial assumptions of the tests used (Shapiro–Wilk test, and Bartlett's test) were checked. The performed tests indicated that the noted data were normally distributed and met additional assumptions for analysis of variance (ANOVA). We used ANOVA and Tukey's post hoc test to compare differences in the attitude scores among categories of respondents, and we used omega square ω^2 to investigate effect size, with cut-off levels of $\omega^2 > 0.01$ for a small effect, $\omega^2 > 0.06$ for a medium effect, and $\omega^2 > 0.14$ for a large effect, as it is widely accepted [37].

3. Results

3.1. Characteristics of Respondents

Among young Polish respondents, the majority were women (52.1%). Approximately half of the respondents invited to these surveys were teenagers living in cities (cities 48%; towns 52%). The majority of respondents were between the ages of 17 and 19 (59%). The remaining respondents (41%) were 14–16 years (38%) or >20 years (3%). More than half of the respondents (53%) visited the urban or managed forest more than once in the

month prior to the survey. The remaining respondents visited the forest once (25%) or not in the last month (22%). Almost half of the respondents brought up in urban areas had both parents who grew up in the countryside (49%). The remaining respondents were raised where their parents grew up in urban areas (37%) or one of them in urban area and the other in the countryside (14%). It turned out that more respondents knew a hunter from their surroundings (48%) than a forest worker (45%). Almost half of the respondents expressed a wish to participate in a hunt as an observer to see what it really looks like (48%). A slightly smaller number did not express such a need (41%) and 11% did not have an opinion. Most of the respondents do not want to be hunters in the future (76%) or have no opinion at present (14%). Only 14% of respondents stated that in the future they would like to become a hunter, and 20% of them were women.

3.2. Attitudes toward Hunting

Half of Polish urban teenagers (51%) recognized that hunting in Poland is necessary to regulate the number of game animals and keep a natural balance in the environment (Table 1). One-third of the respondents (34%) disagreed that populations of wild animals in Poland are currently at a stable level and do not require human regulation. One-fifth of respondents (21%) disagreed that there are fewer and fewer wild animals in Poland because hunters hunt them. The respondents admitted that too little information is communicated by the hunters themselves about what they do and stated that the Polish media show hunting in an objective way. Nearly half of the respondents (49%) disagreed with killing wild animals (by hunters) to obtain meat. Relatively more people would not feel comfortable communicating with the hunter, but a very similar number of people have a different opinion, and one-third did not have an opinion.

The seven questionnaire items (Cronbach's $\alpha = 0.53$) had generally similar factor loadings on principal component 1 (PC1), which was the only component with an eigenvalue > 2 (2.17). PC1 explained 31% of the variance, and thus we judged that it was sufficient to use only this principal component (PC2 = 16.1%). PC1 was highly correlated with the attitude score ($r = 0.9102$, $n = 1947$, $p < 0.001$), which is easier to interpret than PC1. We therefore concluded that the attitude score is a suitable measure of ATH. Many respondents (44.2%) held an ATH that was at least slightly positive (i.e., had an attitude score ≥ 1 ; Figure 1); mean attitude score was 0.17 ($n = 1947$, $SD = 4.47$).

Socio-demographic variables generally had a low impact on attitude toward hunters and hunting (Table 2). Current place of residence and age had no influence on the attitude score ($\omega^2 = 0.00$); however, both were statistically significant. There was a tendency to increase the attitude score level with age ($F = 6.1$; $p = 0.0023$). Only gender was important and appeared to cause medium to small effects ($\omega^2 = 0.00$). Men had on average 3.09 more points on the attitude scale than women.

Forest and hunting experience had a relatively large and significant impact on the attitude score (Table 2). Respondents who expressed their willingness to participate in a hunt as an observer had on average 4.36 more points on the attitude scale than their counterparts ($F = 259.1$; $p < 0.0001$). A similar situation was observed among respondents who expressed their willingness to be a hunter in the future, having on average 4.87 more points on the attitude scale than their counterparts ($F = 185.7$; $p < 0.0001$). Only frequency of visits to the forest, however statistically significant, had no influence on the attitude score ($\omega^2 = 0.12$).

Socialization, i.e., the influence of the conditions their parents grew up in, had a non-statistical impact on attitude toward hunters and hunting (Table 2).

Social network, including knowing any hunter or forest worker, had a small but significant influence on the attitude score (Table 2). Knowing a hunter or forest worker had positive effects on the attitude score ($\omega^2 = 0.05$, and $\omega^2 = 0.04$, respectively). Respondents who know a hunter or forest worker had average attitude scores of 1.93 and 1.72 more points, respectively, than their counterparts.

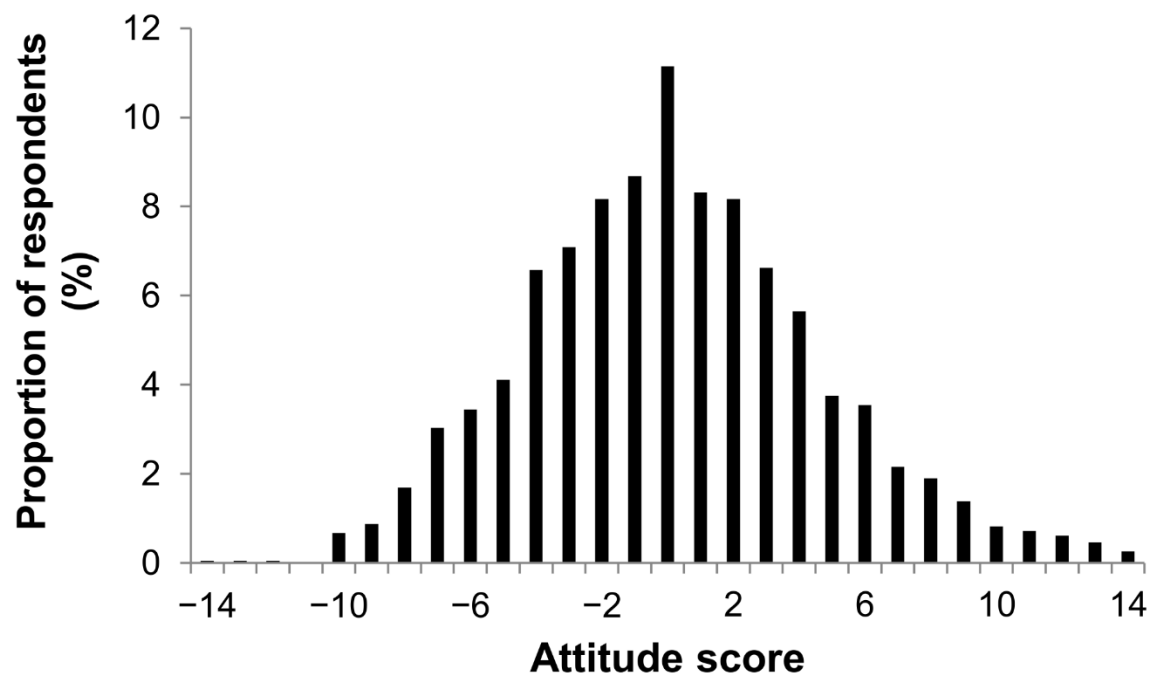


Figure 1. Histogram showing the proportion of young Polish respondents with each attitude score. Attitude score was tallied on seven attitudinal items with respect to hunting and hunters and ranged from -14 to 14 (see Table 1). Data were collected in 2020 from a random sample of 1947 Polish high school students.

Table 2. Average attitude scores of young Polish respondents ($n = 1947$) in relation to their socio-demographic characteristics, forest and hunting experience, socialization, and social network. The attitude score was calculated based on seven attitudinal items, and the attitude scale was centered on 0 (ranging from -14 to 14). Different letters indicate significant differences.

Attitudinal Items (Options)	Attitude Score			<i>p</i> Value	F	ω^2	Effect Size
	First Option	Second Option	Third Option				
Socio-demographic information							
Sex (male; female)	1.79 ± 0.14	−1.30 ± 0.13		<0.0001	264.5	0.12	Medium
Age (14–16; 17–19, >20 years)	−0.17 ± 0.17C	0.34 ± 0.13B	1.72 ± 0.61A	0.0023	6.1	0.00	-
Current place of residence (city; town)	0.43 ± 0.14	−0.06 ± 0.14		0.0155	5.9	0.00	-
Forest and hunting experience							
Frequency of visits to forest (more than once; once; not once)	0.53 ± 0.15A	−0.05 ± 0.19B	−0.42 ± 0.20B	0.0004	7.8	0.00	-
Willingness to participate in the hunt as an observer (yes; no; no opinion)	2.16 ± 0.14A	−2.20 ± 0.14C	0.34 ± 0.22B	<0.0001	259.1	0.21	Large
Willingness to be a hunter in the future (yes; no; no opinion)	4.06 ± 0.31A	−0.81 ± 0.11C	2.69 ± 0.24B	<0.0001	185.7	0.16	Large
Socialization							
Growing up conditions of parents (urban; countryside; urban/countryside)	−0.08 ± 0.16	0.32 ± 0.15	0.37 ± 0.26	0.1493	1.9	0.00	-
Social network							
Knowing any hunter (yes; no)	1.18 ± 0.16	−0.75 ± 0.12		<0.0001	95.9	0.05	Small
Knowing any forest worker (yes; no)	1.12 ± 0.16	−0.60 ± 0.13		<0.0001	74.1	0.04	Small

4. Discussion

Half of Polish urban teenagers recognized that hunting in Poland is a necessity. From socio-demographic factors, forest and hunting experience had the largest impact on ATH.

Gender and social network had a smaller impact but were still significant. Other factors, e.g., the place of residence had no impact on the ATH. The ATH of Polish teenagers is lower than adults [33], and it corresponds with findings about wildlife value orientation. The older people are, the more likely they are to have a utilitarian approach [38].

The leading role of experience in shaping attitude is proved once more [16,33,34], but this time for young people, not only adults. In this context, a law banning children from hunting can have a tremendous impact on attitudes in the future, not only toward hunting but also toward wildlife and nature (including forest) management. There are many examples of successful educational programs that have a big impact on people's values and behaviors e.g., [39,40]. They all were based on Kolb cycle and personal experience as the best didactic approaches, so especially in Poland, forest education based on personal experience is vital for shaping social acceptance of forest management.

Young people ATH was shaped by gender and social network similar to that of adults [16,33]. However, place of residence had no influence on the ATH of Polish adults [33], which is contrary to findings for adults in other European and American countries [14–17,41–44]. Differences can be explained by many different factors, such as forest education programs, history, and culture. but also to differences in survey and research planning and conducting [45].

Our study, although based on a relatively big sample, did not cover the whole country, and it was not random. It was designed as a pilot study before a countrywide survey is planned, so our next step will be a national survey. The change in Polish law to forbid children to hunt is relatively new, so it could not have a big impact on the answers. We believe the results for Poland can still be helpful regarding decision making, not only for Poland but also for other Central European countries.

Our results show that forest education should be focused on all young residents, whether they are coming from a big city, a small town, or a village, and that teenagers' ATH are the same. Field trips and other active methods are recommended concerning forest education because personal experience shapes ATH the most.

Author Contributions: Conceptualization, M.K., A.W. and H.C.; methodology, A.W., A.Ł. and M.K.-P.; formal analysis, A.Ł.; investigation, H.C. and M.K.; resources, M.S.; writing—original draft preparation, H.C. and A.W.; writing—review and editing, H.C., A.W., M.K., A.Ł., M.K.-P. and M.S.; visualization, A.Ł.; supervision, A.W. and M.S.; funding acquisition, M.S. All authors have read and agreed to the published version of the manuscript.

Funding: Publication was financed within the framework of the Polish Ministry of Science and Higher Education's program: "Regional Excellence Initiative" in the years 2019–2022, project no. 005/RID/2018/19.

Data Availability Statement: On request to corresponding author.

Acknowledgments: We want to thank Jolanta Szachniewicz and Damian Cichoń for help in collecting data, and Justin Curtis for linguistic correction.

Conflicts of Interest: The authors declare no conflict of interest.

References

1. FAO. Available online: <http://www.fao.org/forestry/forest-education/en/> (accessed on 30 August 2022).
2. Codrow, H.; Wierzbicka, A.; Skorupski, M. Factors shaping teenagers and young adults' approach to hunting. *Šumarski List* **2023**, 1–2. *in press*.
3. Kowalczyk, A.K.; Borowicz, A.; Gwiazdowicz, D.J. Evaluation of hunting activities by selected social groups. *Acta Sci. Pol. Silv. Colendar. Ratio Ind. Lignar.* **2020**, *19*, 95–103.
4. Matulewska, A.; Gwiazdowicz, D.J. Cyberbullying in Poland: A case study of aggressive messages with emojis targeted at the community of hunters in urbanized society. *Soc. Semiot.* **2020**, *30*, 379–395. [CrossRef]
5. Skogen, K. Who's afraid of the big, bad wolf? Young people's responses to the conflicts over large carnivores in Eastern Norway. *Rural. Sociol.* **2001**, *6*, 203–226. [CrossRef]
6. Wierzbicka, A.; Skorupski, M. What shapes public attitude toward hunting? *Stud. Mater. CEPL* **2017**, *50*, 20–25.
7. Skubis, M.; Skubis, J. Opinion of Poles on hunting and hunters—Survey. *Acta Sci. Pol. Silv. Colendar. Ratio Ind. Lignar.* **2018**, *17*, 163–172.

8. Sobalak, T.; Kapałka-Boratyńska, K.; Wierzbicka, A.; Skorupski, M. No to hunting in Puszcza Zielonka! Case study of a conflict between hunters and members of local community. *Stud. Mater. CEPL* **2017**, *50*, 275–279.
9. Pejnović, D.; Krapinec, K.; Slamar, M. Hunters in Croatia as a socio-geographic group and their socio-demographic characteristics (in Croatian with English summary). *Šumarski List* **2010**, *134*, 461–474.
10. Hauser, P.M. Demographic and ecological changes as factors in outdoor recreation. In *Outdoor Recreation Resources Review Commission Study Report*; Sargent, F.W., Ed.; Government Printing Office: Washington, DC, USA, 1962; pp. 22, 27–59.
11. Kellert, S.R. Attitudes toward animals: Age-related development among children. In *Advances in Animal Welfare Science*; Springer: Berlin/Heidelberg, Germany, 1985; pp. 43–60.
12. Pagani, C.; Robustelli, F.; Ascione, F.R. Italian youths' attitudes towards, and concern for animals. *Anthrozoös* **2007**, *20*, 275–293. [\[CrossRef\]](#)
13. Manfredo, M.J. (Ed.) Who cares about wildlife? In *Who Cares about Wildlife? Social Science Concepts for Exploring Human-Wildlife Relationships and Conservation Issues*; Springer: New York, NY, USA, 2008; pp. 1–27.
14. Butler, J.S.; Shanahan, J.; Decker, D.J. Public attitudes toward wildlife are changing: A trend analysis of New York residents. *Wildl. Soc. B* **2003**, *31*, 1027–1036.
15. MacKay, K.J.; Campbell, J.M. An examination of attitudes toward hunting as a tourism product. *Tour. Manag.* **2004**, *25*, 443–452. [\[CrossRef\]](#)
16. Ljung, P.E.; Riley, S.J.; Heberlein, T.A.; Ericsson, G. Eat Prey and Love: Game-meat consumption and attitudes toward hunting. *Wildl. Soc. B* **2012**, *36*, 669–675. [\[CrossRef\]](#)
17. Sijtsma, M.T.J.; Vaske, J.J.; Jacobs, M.H. Acceptability of lethal control of wildlife that damages agriculture in the Netherlands. *Soc. Natur. Resour.* **2012**, *25*, 1308–1323. [\[CrossRef\]](#)
18. Brämer, R. Das Bambi-Syndrom Naturverklärung als Naturentfremdung. In *Natur Subjektiv—Texte zur Natur-Beziehung in der Hightech-Welt*; Deutsches Wanderinstitut e.V.: Marburg, Germany, 1998; Volume 7, Available online: <http://www.wanderforschung.de/files/bambikz1234003206.pdf> (accessed on 12 July 2022).
19. Kollender, L.; Zabel, J. Nature Experience and Perception of Nature in Peruvian School Students: Closer to Nature, but still far away? In *Strand 9 Environmental, Health and Outdoor Science Education, Proceedings of the ESERA 2017 Conference, Dublin, Ireland, 21–25 August 2017*; ESERA: Dublin, Ireland; pp. 207–217.
20. Smarul, N.; Tomczak, K.; Wierzbicka, A.; Łukowski, A. Possibilities and level of use of Facebook by the State Forests. *Sylvan* **2019**, *163*, 542–550.
21. Krokowska-Paluszak, M.; Wierzbicka, A.; Skorupski, M.; Gruchała, A. Review and analysis of television programs financed by The State Forests National Forest Holding—Their recognition, viewership, and their meaning in The State Forests National Forest Holding brand building and shaping the image of the forester. *Stud. Mater. CEPL* **2016**, *18*, 170–176.
22. Łukowski, A.; Krokowska-Paluszak, M.; Opalińska, P.; Błasiak, A.; Wierzbicka, A.; Skorupski, M.; Sagan, J.; Gruchała, A.; Tomusiak, R. Television as a source of information about game management—Social research. *Stud. Mater. CEPL* **2017**, *5*, 184–191.
23. Wilkins, E.J.; Cole, N.W.; Miller, H.M.; Schuster, R.M.; Dayer, A.A.; Duberstein, J.N.; Fulton, D.C.; Harshaw, H.W.; Raedeke, A.H. Rural-urban differences in hunting and birdwatching attitudes and participation intent. *Hum. Dimens. Wildl.* **2019**, *24*, 530–547. [\[CrossRef\]](#)
24. Aggio, D.; Smith, L.; Fisher, A.; Hamer, M. Mothers' perceived proximity to green space is associated with TV viewing time in children: The Growing Up in Scotland study. *Prev. Med.* **2015**, *70*, 46–49. [\[CrossRef\]](#)
25. Chawla, L.; Derr, V. The Development of Conservation Behaviors in Childhood and Youth. In *Oxford Handbook of Environmental and Conservation Psychology*; Clayton, S.D., Ed.; Oxford University Press: New York, NY, USA, 2012; pp. 527–555.
26. Braun, T.; Dierkes, P. Connecting students to nature—How intensity of nature experience and student age influence the success of outdoor education programs. *Environ. Educ. Res.* **2017**, *23*, 937–949. [\[CrossRef\]](#)
27. Gosling, E.; Williams, K.J.H. Connectedness to nature, place attachment and conservation behaviour: Testing connectedness theory among farmers. *J. Environ. Psychol.* **2010**, *30*, 298–304. [\[CrossRef\]](#)
28. Rosa, C.D.; Profice, C.C.; Collado, S. Nature experiences and adults' self-reported pro-environmental behaviors: The role of connectedness to nature and childhood nature experiences. *Front. Psychol.* **2018**, *9*, 1055. [\[CrossRef\]](#) [\[PubMed\]](#)
29. Gołos, P. Recreational functions of Warsaw's urban and suburban forests. *For. Res. Pap.* **2013**, *74*, 57–70.
30. Małek, J. Historical and contemporary conditioning of suburbanization processes. *Przestrz. i Forma* **2011**, *16*, 431–442.
31. Markuszewska, I.; Delebis, A. Urbanization of rural areas in the perception of local residents on the example of Wolica near Kalisz. *Bad. Fizjogr.* **2016**, *67*, 145–154.
32. Woods, M. Deconstructing rural protest: The emergence of a new social movement. *J. Rural. Stud.* **2003**, *19*, 309–325. [\[CrossRef\]](#)
33. Krokowska-Paluszak, M.; Łukowski, A.; Wierzbicka, A.; Gruchała, A.; Sagan, J.; Skorupski, M. Attitudes towards hunting in Polish society and the related impacts of hunting experience, socialisation and social networks. *Eur. J. Wildl. Res.* **2020**, *66*, 73. [\[CrossRef\]](#)
34. Krokowska-Paluszak, M.; Wierzbicka, A.; Łukowski, A.; Gruchała, A.; Sagan, J.; Skorupski, M. Attitudes towards Foresters in Polish Society. *Forests* **2022**, *13*, 294. [\[CrossRef\]](#)
35. Haligowski, W. The issue of constitutionality of the prohibition to hunt in the presence or with participation of children. In *Prawo Wobec Wyzwań Współczesności: Z Zagadnień Nauk Penalnych*; Helios, J., Jedlecka, W., Kwieciński, A., Eds.; Uniwersytet Wrocławski: Wrocław, Poland, 2019; Volume 145, pp. 95–106.

36. Vaske, J. *Survey Research and Analysis: Applications in Parks, Recreation and Human Dimensions*; Venture Publishing: State College, PA, USA, 2008.
37. Field, A.P. *Discovering Statistics with IBM SPSS Statistics*, 4th ed.; SAGE: Thousand Oaks, CA, USA, 2013.
38. Fulton, D.C.; Manfredo, M.J.; Lipscomb, J. Wildlife value orientations: A conceptual and measurement approach. *Hum. Dimens. Wildl.* **2008**, *1*, 24–47. [[CrossRef](#)]
39. Shapiro, H.G.; Erickson, K.A.; Nils Peterson, M.; Frew, K.N.; Stevenson, K.T.; Langerhaus, R.B. Which species to conserve: Evaluating children's species-based conservation priorities. *Biodivers. Conserv.* **2016**, *25*, 539–553. [[CrossRef](#)]
40. Sponarski, C.C.; Vaske, J.J.; Bath, A.J.; Loeffler, T.A. Changing attitudes and emotions toward coyotes with experimental education. *J. Environ. Educ.* **2016**, *47*, 296–306. [[CrossRef](#)]
41. Heberlein, T.A.; Ericsson, G. Ties to the countryside: Accounting for urbanites attitudes toward hunting wolves and wildlife. *Hum. Dimens. Wildl.* **2005**, *10*, 213–227. [[CrossRef](#)]
42. Peterson, M.N.; Mertig, A.G.; Liu, J. Effects of zoonotic disease attributes on public attitudes towards wildlife management. *J. Wildl. Manag.* **2006**, *70*, 1746–1753. [[CrossRef](#)]
43. Gangaas, K.E.; Kaltenborn, B.P.; Andreassen, H.P. mGeo-spatial aspects of acceptance of illegal hunting of large carnivores in Scandinavia. *PLoS ONE* **2013**, *8*, e68849. [[CrossRef](#)]
44. Ljung, P.E.; Riley, S.J.; Ericsson, G. Game meat consumption feeds urban support of traditional use of natural resources. *Soc. Natur. Resour.* **2015**, *28*, 657–669. [[CrossRef](#)]
45. Krokowska-Paluszak, M.; Łukowski, A.; Gruchała, A.; Skorupski, M. Las i leśnicy oczami społeczeństwa—Co wpływa na wizerunek Lasów Państwowych i leśników—Badania sondażowe w województwie wielkopolskim. In Proceedings of the Zimowa Szkoła Leśna. X Sesja Współczesne Problemy Komunikacji Społecznej i Edukacji w Leśnictwie, Sękocin Stary, Poland, 13–15 March 2018; pp. 323–343.